**Skills Review**

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**Course/Section: CPE4B** **Date: Oct. 7, 2026**

**Example No. 13:** This program will prompt the user to press uppercase or lowercase letters from the keyboard and respond with a message of what letter has been pressed by the user. The program uses the function 07h to facilitate input without echo. Save the program using sample13.asm as the filename

**Program**

```
; AARON LUDWIG A. ALTAR
; CPE4B

.model small
.stack
.data

p0      DB 'Press UPPERCASE or lowercase letter  A/a     ',13,10,'$'
p1      DB 'You Pressed UPPERCASE letter A           ',13,10,'$'
p2      DB 'You Pressed LOWERCASE letter a            ',13,10,'$'
p3      DB 'Press any Key to continue [X] to Exit ',13,10,'$'
p4      DB '
',13,10,'$'

.code
start:
        mov ax, @data           ; initialize the data segment
        mov ds, ax

        mov ah, 09h             ; Function 09h - display string
        mov dx, offset p0       ; prompt: press A or a
        int 21h
        mov dx, offset p4       ; blank line
        int 21h

        mov ah, 07h             ; Function 07h - read a key without echo
        int 21h                 ; AL = key pressed

        cmp al, 41h             ; is it 'A' (41h)?
        je Malaki               ; if so, go to Malaki (uppercase)
        cmp al, 61h             ; is it 'a' (61h)?
        je Maliit               ; if so, go to Maliit (lowercase)
        jne start               ; any other key - ask again
```

```
Malaki:
        mov ah, 09h
        mov dx, offset p1       ; "You Pressed UPPERCASE letter A"
        int 21h

        mov dx, offset p3       ; continue or exit prompt
        int 21h

        mov ah, 07h             ; wait for a key without echo
        int 21h

        cmp al, 'X'             ; is it X?
        je EXIT                 ; if so, end the program
        cmp al, 'x'             ; also accept lowercase x
        je EXIT
        jne start               ; otherwise, start again

Maliit:
        mov ah, 09h
        mov dx, offset p2       ; "You Pressed LOWERCASE letter a"
        int 21h

        mov dx, offset p3       ; continue or exit prompt
        int 21h

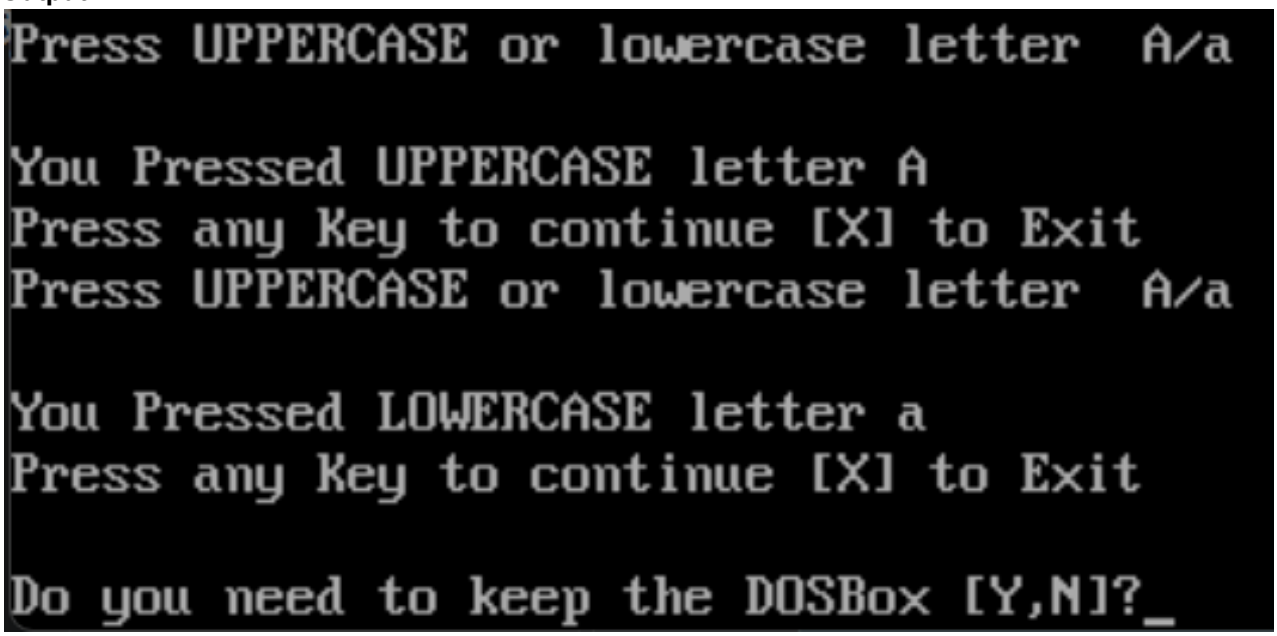
        mov ah, 07h             ; wait for a key without echo
        int 21h

        cmp al, 'X'             ; is it X?
        je EXIT                 ; if so, end the program
        cmp al, 'x'             ; also accept lowercase x
        je EXIT
        jne start               ; otherwise, start again

EXIT:   mov ah, 4Ch             ; terminate program
        int 21h

end start
```

**Output**


Press UPPERCASE or lowercase letter  A/a
You Pressed UPPERCASE letter A
Press any Key to continue [X] to Exit
Press UPPERCASE or lowercase letter  A/a
You Pressed LOWERCASE letter a
Press any Key to continue [X] to Exit
Do you need to keep the DOSBox [Y,N]?_